Applications of Corner and Shape Features in Digital Image Processing: Three Case Studies

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***Abstract*—Digital image processing supports the identification of meaningful objects and structures from visual data. However, different problems require different image features to obtain an appropriate result. This report reviews three published case studies: highway lane detection using the Hough Transform, large aerial image formation using Harris–Plessey corner detection, and red blood cell counting using the Circular Hough Transform. Each study is examined based on its problem, processing method, suitable features, and experimental findings. The comparison shows that line features are suitable for road boundaries, corner features are useful for matching overlapping images, and circular contours are appropriate for identifying blood cells. Therefore, feature selection should be determined by the geometric characteristics of the target object and the objective of image analysis.**

***Index Terms*—Digital image processing, feature extraction, Harris corner detector, Hough Transform, image stitching, lane detection, blood cell counting.**

I. Introduction

Nowadays, digital image processing is implemented in various fields, including transportation, mapping, and healthcare. It allows a computer system to analyze images and identify information that may be difficult to process manually. However, the objects contained in an image have different characteristics. A highway lane can be recognized through its line structure, while aerial image fragments require distinctive points to identify overlapping regions. Meanwhile, blood cells are commonly identified through their approximately circular boundaries.

Feature extraction is an important step because it converts visual characteristics into information that can be used for detection, matching, or measurement. Corner detection focuses on locations where image intensity changes in multiple directions, whereas the Hough Transform identifies geometric structures by accumulating votes for possible shape parameters. This report discusses three published implementations of these techniques and identifies which features are suitable for each problem. The discussion focuses on reviewing existing research rather than conducting new experiments.

II. Feature Extraction Techniques

A. Harris–Plessey Corner Detection

The Harris corner detector identifies image points with significant intensity changes in more than one direction. These points are relatively distinctive compared with flat regions or simple edges. The detector evaluates a local image-gradient matrix and produces a corner response, commonly represented as

$$R = \det(M) - k\big(\mathrm{trace}(M)\big)^2, \tag{1}$$

where M is the local second-moment matrix and k is a detector constant. Large positive responses indicate potential corner points. For image matching, the selected points should also be sufficiently repeatable across overlapping frames.

B. Hough Transform and Circular Hough Transform

The Hough Transform detects shapes by allowing feature points, typically edge pixels, to vote for geometric parameters. For straight-line detection, a common representation is

$$\rho = x\cos\theta + y\sin\theta, \tag{2}$$

where ρ is the perpendicular distance from the origin and θ is the line's normal angle. A high number of votes indicates a likely line. The Circular Hough Transform extends the same principle to circles, described by

$$(x-a)^2 + (y-b)^2 = r^2, \tag{3}$$

where (a, b) is the circle center and r is its radius. These techniques are suitable for objects with recognizable straight or circular boundaries.

III. Case Study Analysis

A. Case 1: Highway Lane Detection

1) Problem and Objective: Aminuddin *et al.* [1] developed a highway lane detection algorithm using videos recorded by an on-board vehicle camera in Malaysia. Road environments contain other objects, such as trees, road dividers, buildings, and vehicles, which can interfere with lane detection. Changes in weather and illumination also make the detection process more difficult. The study aimed to detect lane markings to support vanishing-point detection, road-width measurements, and identification of the road region of interest (ROI).

2) Implementation and Suitable Features: The algorithm first reads image frames and performs image enhancement and smoothing to reduce noise. It then applies the Sobel operator to extract edges before using the Hough Transform to detect the lane lines. The detected left and right lane boundaries support geometric calculations for the vanishing point, road width, and ROI.

The most suitable features are *edge pixels, straight-line segments, line orientation, and the intersection geometry of lane boundaries*. These features are appropriate because lane markings often appear as elongated, high-contrast structures against the road surface. In this case, the Hough Transform helps group edge evidence into meaningful lines instead of treating individual edge pixels as separate objects.

3) Reported Results and Limitations: The researchers evaluated 6,469 frames from normal, rainy, night-time, and rural-road scenes. Their reported average true-positive rate was 0.925, with an average false-negative rate of 0.075 and false-positive rate of 0.129 [1]. The paper also notes that faded or obscured markings and background edges resembling lanes can cause detection errors. Therefore, line extraction is effective for this task, but its performance still depends on road conditions and the visibility of lane markings.

B. Case 2: Large Aerial Image Formation

1) Problem and Objective: Mateika and Martavičius [2] investigated the formation of large aerial images by stitching multiple smaller image fragments. Aerial photographs may contain buildings, forests, or other changing scenery, and they cannot reliably depend on artificial reference markers. For accurate stitching, corresponding positions in overlapping fragments must first be identified.

2) Implementation and Suitable Features: The authors proposed an automated photomap stitching method based on Harris–Plessey corner detection. Their process calculates a cornerness map, applies thresholding to isolate promising corner regions, and selects local maxima as corner points. These distinctive points can then support matching between overlapping aerial image fragments. The paper places particular emphasis on controlling the number of detected corners through the threshold chosen from image statistics.

The suitable features are *high-response corner points, their spatial locations, and repeatable keypoints within overlapping regions*. Corner points are appropriate because they provide more distinctive two-dimensional locations than uniform areas or isolated straight edges. Such features help determine which locations in one image correspond to locations in another.

3) Reported Results and Limitations: The study tested seven overlapping aerial-image samples whose content gradually changed from a highly detailed townscape to woodland. A fixed threshold produced corner-point counts that varied by more than a factor of two, whereas histogram-guided threshold selection reduced the variation to less than 20% in the reported experiment [2]. The findings support adaptive corner selection, although successful matching also requires sufficient overlap and appropriate image content. The paper recommends multiple overlapping fragments and discusses overlap greater than 50% for reliable registration.

C. Case 3: Red Blood Cell Counting

1) Problem and Objective: Bagui and Zoueu [3] studied automatic red blood cell counting from multispectral microscopic images. Manual counting can require considerable laboratory time, and automated recognition is challenging when cells overlap, differ in size, or contain noise. The objective was to detect and count red blood cells using digital image processing to support microscopy analysis.

2) Implementation and Suitable Features: The proposed method processes multispectral images, applies preprocessing and Canny edge detection to individual spectral images, and combines the resulting edge information. The Circular Hough Transform is then used to find circular structures, followed by segmentation and counting. The researchers consider a range of possible cell radii instead of assuming that every cell has the same size.

The most suitable features are *circular cell contours, edge gradients, candidate circle centers, and cell radius*. These features are suitable because red blood cells often appear approximately circular under the microscope. Compared with a straight-line detector, the Circular Hough Transform more directly represents the expected cell geometry and can use accumulated evidence from boundary pixels.

3) Reported Results and Limitations: The researchers evaluated 25 blood samples and reported a counting rate of 98.07%, corresponding to a mean counting error of 1.93% compared with manual counting [3]. The study also discusses complications caused by overlapping cells and imperfect boundaries. These results demonstrate the usefulness of circular shape features, while indicating that segmentation quality and the assumed radius range influence cell detection.

IV. Comparison and Discussion

The three case studies show that the most suitable feature depends on the problem being solved. Although highway lane detection and red blood cell counting both use the Hough Transform, they require different geometric models. Straight-line parameters are appropriate for lanes, whereas circle centers and radii are needed for blood cells. Aerial image stitching differs because its main goal is to find corresponding locations between images rather than to detect a particular geometric object.

The studies also show that extracting a feature is not sufficient by itself. Preprocessing and parameter selection affect the final result. Lane detection can be affected by lighting and misleading background edges; aerial stitching depends on corner thresholds and overlap; blood cell counting depends on clear boundaries and an appropriate radius range. Consequently, feature extraction methods should be chosen based on both the target's shape and the conditions in which images are captured.

TABLE I
COMPARISON OF THE THREE DIGITAL IMAGE PROCESSING CASE STUDIES

Application	**Main Technique**	**Suitable Image Features**	**Reason for Feature Selection**
Highway lane detection	Sobel edge detection and Hough Transform	Edges, straight lines, orientation, intersection geometry	Lane markings can be modeled as elongated line boundaries for locating the road region.
Aerial image stitching	Harris–Plessey corner detection	Corner responses, keypoint positions, overlapping correspondences	Distinctive and repeatable corner points can be matched between image fragments.
Red blood cell counting	Canny edge detection and Circular Hough Transform	Circular contours, circle centers, radii	Cell outlines are often approximately circular and can be identified through boundary votes.

V. CONCLUSION

This report reviewed three real-world implementations of digital image processing and identified the image features most suitable for each case. Highway lane detection benefits from edge and line features extracted with the Hough Transform. Aerial image stitching uses distinctive Harris–Plessey corner points to locate overlapping regions, while red blood cell counting relies on circular contours extracted with the Circular Hough Transform. These examples demonstrate that different visual problems require different feature representations. Choosing features that correspond to the target object's geometry, together with suitable preprocessing, is important for obtaining reliable digital image processing results.

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